

1. Group the given data (185, 72), (170, 56), (168, 60), (179, 68), (182, 72), (188, 77) into two groups using an unsupervised algorithm up to two iterations using euclidean distance . Initially choose the first two objects as initial centroids. [15 marks].

Instance	X	Y
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77

Solution:

Given, number of clusters to be created (K) = 2 say c1 and c2,
number of iterations = 2 and The given data points can be represented in tabular form as:also, first two objects as initial centroids:Centroid for first cluster c1 = (185, 72)Centroid for second cluster c2 = (170, 56)

Iteration 1: Now calculating similarity by using *Euclidean distance* measure as: Euclidean distance calculation

$$d(c1, 3) = \sqrt{(185 - 168)^2 + (72 - 60)^2} = \sqrt{(17)^2 + (12)^2} = \sqrt{289 + 144} = \sqrt{433}$$

$$d(c2, 3) = \sqrt{(170 - 168)^2 + (56 - 60)^2} = \sqrt{(2)^2 + (-4)^2} = \sqrt{4 + 16} = \sqrt{20}$$

Here, $d(c2, 3) < d(c1, 3)$

So, data point 3 belongs to c2.

$$d(c1, 4) = \sqrt{(185 - 179)^2 + (72 - 68)^2} = \sqrt{(6)^2 + (4)^2} = \sqrt{36 + 16} = \sqrt{52}$$

$$d(c2, 4) = \sqrt{(170 - 179)^2 + (56 - 68)^2} = \sqrt{(-9)^2 + (-12)^2} = \sqrt{81 + 144} = \sqrt{225}$$

Here, $d(c1, 4) < d(c2, 4)$

So, data point 4 belongs to c1.

$$d(c1, 5) = \sqrt{(185 - 182)^2 + (72 - 72)^2} = \sqrt{(3)^2 + (0)^2} = \sqrt{9}$$

$$d(c2, 5) = \sqrt{(170 - 182)^2 + (56 - 72)^2} = \sqrt{(-12)^2 + (-16)^2} = \sqrt{144 + 256} = \sqrt{400}$$

Here, $d(c1, 5) < d(c2, 5)$

So, data point 5 belongs to c1.

$$d(c1, 6) = \sqrt{(185 - 188)^2 + (72 - 77)^2} = \sqrt{(-3)^2 + (-5)^2} = \sqrt{9 + 25} = \sqrt{34}$$

$$d(c2, 6) = \sqrt{(170 - 188)^2 + (56 - 77)^2} = \sqrt{(-18)^2 + (-21)^2} = \sqrt{324 + 441} = \sqrt{765}$$

Here, $d(c1, 6) < d(c2, 6)$

So, data point 6 belongs to c1.

Instance	X	Y	Distance(C1)	Distance(C2)	Cluster
1	185	72			c1
2	170	56			c2
3	168	60	$\sqrt{433}$	$\sqrt{20}$	c2
4	179	68	$\sqrt{52}$	$\sqrt{225}$	c1
5	182	72	$\sqrt{9}$	$\sqrt{400}$	c1
6	188	77	$\sqrt{34}$	$\sqrt{765}$	c1

Representing above information in tabular form:

Distance of each data points from cluster centroids

The resulting cluster after first iteration is: C1- 1 4 5 6, C2 2 3

Iteration 2: Now calculating centroid for each cluster

$$\text{Centroid for first cluster } c1 = \left(\frac{185+179+182+188}{4}, \frac{72+68+72+77}{4} \right) = (183.5, 72.25)$$

$$\text{Centroid for second cluster } c2 = \left(\frac{170+168}{2}, \frac{56+60}{2} \right) = (169, 58)$$

Calculating centroid as mean of data points Now, again calculating similarity:

$$d(c1, 1) = \sqrt{(183.5 - 185)^2 + (72.25 - 72)^2} = 1.5207$$

$$d(c2, 1) = \sqrt{(169 - 185)^2 + (58 - 72)^2} = 21.2603$$

Here, $d(c1, 1) < d(c2, 1)$

So, data point 1 belongs to c1.

$$d(c1, 2) = \sqrt{(183.5 - 170)^2 + (72.25 - 56)^2} = 21.1261$$

$$d(c2, 2) = \sqrt{(169 - 170)^2 + (58 - 56)^2} = 2.2361$$

Here, $d(c2, 2) < d(c1, 2)$

So, data point 2 belongs to c2.

$$d(c1, 3) = \sqrt{(183.5 - 168)^2 + (72.25 - 60)^2} = 19.7563$$

$$d(c2, 3) = \sqrt{(169 - 168)^2 + (58 - 60)^2} = 2.2361$$

Here, $d(c2, 3) < d(c1, 3)$

So, data point 3 belongs to c2.

$$d(c1, 4) = \sqrt{(183.5 - 179)^2 + (72.25 - 68)^2} = 6.1897$$

$$d(c2, 4) = \sqrt{(169 - 179)^2 + (58 - 68)^2} = 14.1421$$

Here, $d(c1, 4) < d(c2, 4)$

So, data point 4 belongs to c1.

$$d(c1, 5) = \sqrt{(183.5 - 182)^2 + (72.25 - 72)^2} = 1.5207$$

$$d(c2, 5) = \sqrt{(169 - 182)^2 + (58 - 72)^2} = 19.1050$$

Here, $d(c1, 5) < d(c2, 5)$

So, data point 5 belongs to c1.

$$d(c1, 6) = \sqrt{(183.5 - 188)^2 + (72.25 - 77)^2} = 6.5431$$

$$d(c2, 6) = \sqrt{(169 - 188)^2 + (58 - 77)^2} = 26.8701$$

Here, $d(c1, 6) < d(c2, 6)$

So, data point 6 belongs to c1.

Distance calculation between data points and centroids

Representing above information in tabular form.

Instance	X	Y	Distance(C1)	Distance(C2)	Cluster
1	185	72	1.5207	21.2603	c1
2	170	56	21.1261	2.2361	c2
3	168	60	19.7563	2.2361	c2
4	179	68	6.1897	14.1421	c1
5	182	72	1.5207	19.105	c1
6	188	77	6.5431	26.8701	c1

Distance of each data points from cluster centroids

The resulting cluster after second iteration is: C1- 1 4 5 6, C2= 2 3

- For the given data from a retail company predict monthly sales revenue for size 2800 sq ft and 12 employees using KNN based on three nearest neighbors using Euclidean distance.. **[10 marks]**

Store Size (sq ft)	Number of Employees	Monthly Sales Revenue (\$)
1500	5	30000
2000	7	50000
2500	10	70000
3000	15	80000
3500	20	100000

the

Answer:

Calculate Distances:

- **For Store 1 (1500 sq ft, 5 employees):** Distance = $\sqrt{(1500 - 2800)^2 + (5 - 12)^2}$
Distance = $\sqrt{(1300)^2 + (-7)^2}$
Distance = $\sqrt{1690000 + 49}$
Distance ≈ 1300.02
- **For Store 2 (2000 sq ft, 7 employees):** Distance = $\sqrt{(2000 - 2800)^2 + (7 - 12)^2}$
Distance = $\sqrt{(800)^2 + (-5)^2}$
Distance = $\sqrt{640000 + 25}$
Distance ≈ 800.02
- **For Store 3 (2500 sq ft, 10 employees):** Distance = $\sqrt{(2500 - 2800)^2 + (10 - 12)^2}$
Distance = $\sqrt{(-300)^2 + (-2)^2}$
Distance = $\sqrt{90000 + 4}$
Distance ≈ 300.01
- **For Store 4 (3000 sq ft, 15 employees):** Distance = $\sqrt{(3000 - 2800)^2 + (15 - 12)^2}$
Distance = $\sqrt{(200)^2 + (3)^2}$
Distance = $\sqrt{40000 + 9}$
Distance ≈ 200.02
- **For Store 5 (3500 sq ft, 20 employees):** Distance = $\sqrt{(3500 - 2800)^2 + (20 - 12)^2}$
Distance = $\sqrt{(700)^2 + (8)^2}$
Distance = $\sqrt{490000 + 64}$
Distance ≈ 700.04

Step 4: Finding Neighbors

Now, we have calculated the distances for each store. Here are the distances we obtained:

- Store 1: 1300.02
- Store 2: 800.02
- Store 3: 300.01
- Store 4: 200.02
- Store 5: 700.04

Now, we will select the three closest neighbors (smallest distances):

1. Store 4 (200.02)
2. Store 3 (300.01)
3. Store 5 (700.04)

Step 5: Target Value Prediction

Next, we will take the monthly sales revenues of these three nearest neighbors and calculate their average.

- Store 4 Revenue: \$80,000
- Store 3 Revenue: \$70,000
- Store 5 Revenue: \$100,000

Average Revenue = $(80000 + 70000 + 100000) / 3$

Average Revenue = $250000 / 3$

Average Revenue ≈ 83333.33

The predicted monthly sales revenue for the new store (2800 sq ft and 12 employees) is approximately **\$83,333.33**.

3. (a) Build a decision tree using the ID3 algorithm for the given training data in the table (Buy Computer data), and predict the class of the following new example: age $\leq$ 30, income=medium, student=yes, credit-rating=fair. [20 marks].

age	income	student	Credit rating	Buys computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31...40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
> 40	medium	no	excellent	no

The mutual information of the two classes,

$$\text{Entropy}(S) = E(9,5) = -9/14 \log_2(9/14) - 5/14 \log_2(5/14) = 0.94$$

Now Consider the Age attribute

For Age, we have three values $\text{age}_{<=30}$ (2 yes and 3 no), $\text{age}_{31..40}$ (4 yes and 0 no), and $\text{age}_{>40}$ (3 yes and 2 no)

$$\text{Entropy}(\text{age}) = 5/14 (-2/5 \log_2(2/5) - 3/5 \log_2(3/5)) + 4/14 (0) + 5/14 (-3/5 \log_2(3/5) - 2/5 \log_2(2/5))$$

$$= 5/14(0.9709) + 0 + 5/14(0.9709) = 0.6935$$

$$\text{Gain}(\text{age}) = 0.94 - 0.6935 = 0.2465$$

Next, consider Income Attribute

For Income, we have three values $\text{income}_{\text{high}}$ (2 yes and 2 no), $\text{income}_{\text{medium}}$ (4 yes and 2 no), and $\text{income}_{\text{low}}$ (3 yes 1 no)

$$\text{Entropy}(\text{income}) = 4/14 (-2/4 \log_2(2/4) - 2/4 \log_2(2/4)) + 6/14 (-4/6 \log_2(4/6) - 2/6 \log_2(2/6)) + 4/14 (-3/4 \log_2(3/4) - 1/4 \log_2(1/4))$$

$$= 4/14 (1) + 6/14 (0.918) + 4/14 (0.811)$$

$$= 0.285714 + 0.393428 + 0.231714 = 0.9108$$

$$\text{Gain}(\text{income}) = 0.94 - 0.9108 = 0.0292$$

Next, consider Student Attribute

For Student, we have two values $\text{student}_{\text{yes}}$ (6 yes and 1 no) and $\text{student}_{\text{no}}$ (3 yes 4 no)

$$= 4/14 (1) + 6/14 (0.918) + 4/14 (0.811)$$

$$= 0.285714 + 0.393428 + 0.231714 = 0.9108$$

$$\text{Gain}(\text{income}) = 0.94 - 0.9108 = 0.0292$$

Next, consider Student Attribute

For Student, we have two values $\text{student}_{\text{yes}}$ (6 yes and 1 no) and $\text{student}_{\text{no}}$ (3 yes 4 no)

$$\text{Entropy}(\text{student}) = 7/14(-6/7\log_2(6/7) - 1/7\log_2(1/7)) + 7/14(-3/7\log_2(3/7) - 4/7\log_2(4/7))$$

$$= 7/14(0.5916) + 7/14(0.9852)$$

$$= 0.2958 + 0.4926 = 0.7884$$

$$\text{Gain}(\text{student}) = 0.94 - 0.7884 = 0.1516$$

Finally, consider Credit_Rating Attribute

For Credit_Rating we have two values $\text{credit_rating}_{\text{fair}}$ (6 yes and 2 no) and $\text{credit_rating}_{\text{excellent}}$ (3 yes 3 no)

$$\text{Entropy}(\text{credit_rating}) = 8/14(-6/8\log_2(6/8) - 2/8\log_2(2/8)) + 6/14(-3/6\log_2(3/6) - 3/6\log_2(3/6))$$

$$= 8/14(0.8112) + 6/14(1)$$

$$= 0.4635 + 0.4285 = 0.8920$$

$$\text{Gain}(\text{credit_rating}) = 0.94 - 0.8920 = \mathbf{0.048}$$

Since Age has the highest Information Gain we start splitting the dataset using the age attribute.

The same process of splitting has to happen for the two remaining branches.

Income	student	credit_rating	Class
high	no	fair	No
high	no	excellent	No
medium	no	fair	No
low	yes	fair	Yes
medium	yes	excellent	Yes

Left sub-branch

For branch $age \leq 30$ we still have attributes income, student, and credit_rating. Which one should be used to split the partition?

The mutual information is $E(S_{age \leq 30}) = E(2,3) = -2/5 \log_2(2/5) - 3/5 \log_2(3/5) = 0.97$

For **Income**, we have three values $income_{high}$ (0 yes and 2 no), $income_{medium}$ (1 yes and 1 no) and $income_{low}$ (1 yes and 0 no)

$Gain(income) = 0.97 - 0.4 = 0.57$

For Student, we have two values $student_{yes}$ (2 yes and 0 no) and $student_{no}$ (0 yes 3 no)

$Entropy(student) = 2/5(0) + 3/5(0) = 0$

$Gain(student) = 0.97 - 0 = 0.97$

We can then safely split on attribute student without checking the other attributes since the information gain is maximized.

Now build the decision tree for right left subtree

Income	student	credit_rating	Class
medium	no	fair	Yes
low	yes	fair	Yes
low	yes	excellent	No
medium	yes	fair	Yes
medium	no	excellent	No

Right sub-branch

The mutual information is $Entropy(S_{age > 40}) = I(3,2) = -3/5 \log_2(3/5) - 2/5 \log_2(2/5) = 0.97$

For **Income**, we have two values $\text{income}_{\text{medium}}$ (2 yes and 1 no) and $\text{income}_{\text{low}}$ (1 yes and 1 no)

$$\text{Entropy}(\text{income}) = \frac{3}{5}(-\frac{2}{3}\log_2(\frac{2}{3}) - \frac{1}{3}\log_2(\frac{1}{3})) + \frac{2}{5}(-\frac{1}{2}\log_2(\frac{1}{2}) - \frac{1}{2}\log_2(\frac{1}{2}))$$

$$= \frac{3}{5}(0.9182) + \frac{2}{5}(1) = 0.55 + 0.4 = 0.95$$

$$\text{Gain}(\text{income}) = 0.97 - 0.95 = 0.02$$

For **Student**, we have two values $\text{student}_{\text{yes}}$ (2 yes and 1 no) and $\text{student}_{\text{no}}$ (1 yes and 1 no)

$$\text{Entropy}(\text{student}) = \frac{3}{5}(-\frac{2}{3}\log_2(\frac{2}{3}) - \frac{1}{3}\log_2(\frac{1}{3})) + \frac{2}{5}(-\frac{1}{2}\log_2(\frac{1}{2}) - \frac{1}{2}\log_2(\frac{1}{2})) = 0.95$$

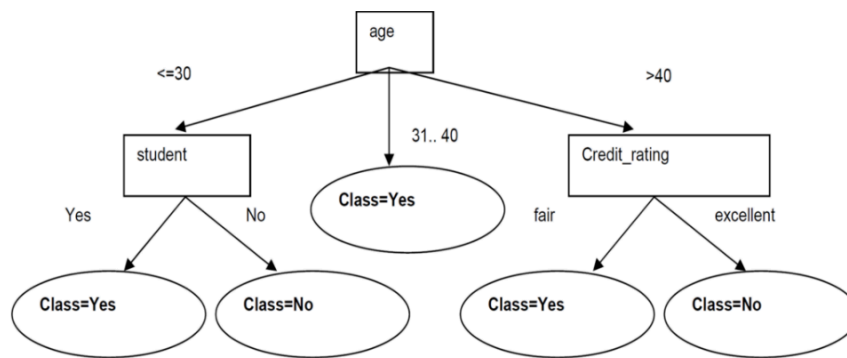
$$\text{Gain}(\text{student}) = 0.97 - 0.95 = 0.02$$

For **Credit_Rating**, we have two values $\text{credit_rating}_{\text{fair}}$ (3 yes and 0 no) and $\text{credit_rating}_{\text{excellent}}$ (0 yes and 2 no)

$$\text{Entropy}(\text{credit_rating}) = 0$$

$$\text{Gain}(\text{credit_rating}) = 0.97 - 0 = 0.97$$

We then split based on **credit_rating**. These splits give partitions each with records from the same class. We just need to make these into leaf nodes with their class label attached:



Decision Tree for Buys Computer

New example: age<=30, income=medium, student=yes, credit-rating=fair

Follow branch(age<=30) then student=yes we predict Class=yes

Buys_computer = yes

(b) For the given confusion matrix compute accuracy, precision and recall [1+2+2=5 marks]

$$\text{Accuracy} = \frac{57+102}{7+5+57+102} = 92.98\%$$

$$\text{Precision} = \frac{57}{57+5} = 0.9193$$

$$\text{Recall} = \frac{57}{57+7} = 0.8906$$

Confusion Matrix

	Actual 0	Actual 1
Predicted 0	57	7
Predicted 1	5	102

The confusion matrix is a 2x2 grid. The top row represents predicted class 0, with a light blue cell for 57 true positives and a light gray cell for 7 false positives. The bottom row represents predicted class 1, with a light gray cell for 5 false negatives and a dark blue cell for 102 true negatives. The axes are labeled 'Actuals' and 'Predictions'.

4. A company wants to classify customer feedback as either "Positive" or "Negative" based on two features: "Sentiment Score" (continuous, from -1 to 1) and "Contains Keyword" (categorical, Yes/No) with given training data. For the given new data point of a new customer feedback has a "Sentiment Score" of 0.7 and "Contains Keyword" as "No". Classify this feedback using Naive Bayes. [10 marks]

Sentiment Score	Contains Keyword	Class
0.8	Yes	Positive
0.6	Yes	Positive
0.9	No	Positive
-0.2	Yes	Negative
-0.7	No	Negative
-0.5	No	Negative

STEP 1 — Compute Prior Probabilities

We have 6 training samples:

- Positive = 3
- Negative = 3

So:

$$P(\text{Positive}) = \frac{3}{6} = 0.5$$

$$P(\text{Negative}) = \frac{3}{6} = 0.5$$

STEP 2 — Compute Likelihoods for "Contains Keyword"

For the Positive class:

Positive samples:

- Yes
- Yes
- No

$$P(\text{No} | \text{Positive}) = \frac{1}{3}$$

For the Negative class:

Negative samples:

- Yes
- No
- No

$$P(\text{No} | \text{Negative}) = \frac{2}{3}$$

STEP 3 — Compute Gaussian Likelihood for Sentiment Score

We need means and variances.

Positive class scores:

0.8, 0.6, 0.9

Mean:

$$\mu_+ = \frac{0.8 + 0.6 + 0.9}{3} = \frac{2.3}{3} = 0.7667$$

Variance:

$$\sigma_+^2 = \frac{(0.8 - 0.7667)^2 + (0.6 - 0.7667)^2 + (0.9 - 0.7667)^2}{3}$$

Compute differences:

- $(0.8 - 0.7667)^2 = 0.0011$
 - $(0.6 - 0.7667)^2 = 0.0278$
 - $(0.9 - 0.7667)^2 = 0.0178$
-

Sum = 0.0467

$$\sigma_+^2 = \frac{0.0467}{3} = 0.01556$$

Negative class scores:

-0.2, -0.7, -0.5

Mean:

$$\mu_- = \frac{-0.2 - 0.7 - 0.5}{3} = \frac{-1.4}{3} = -0.4667$$

Variance:

- $(-0.2 + 0.4667)^2 = 0.0711$
 - $(-0.7 + 0.4667)^2 = 0.0544$
-

- $(-0.5 + 0.4667)^2 = 0.0011$

Sum = 0.1266

$$\sigma_-^2 = \frac{0.1266}{3} = 0.0422$$

STEP 4 — Compute Gaussian likelihoods

The Gaussian formula:

$$P(x | C) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

For Positive class ($x = 0.7$)

$$(x - \mu_+)^2 = (0.7 - 0.7667)^2 = 0.0044$$

$$P(0.7 | +) = \frac{1}{\sqrt{2\pi(0.01556)}} \exp\left(-\frac{0.0044}{2(0.01556)}\right)$$

Compute exponent:

$$\frac{0.0044}{0.03112} \approx 0.141$$

$$e^{-0.141} \approx 0.868$$

Compute denominator:

$$\sqrt{2\pi(0.01556)} \approx 0.312$$

So:

$$P(0.7 | +) = \frac{0.868}{0.312} \approx 2.78$$

For Negative class ($x = 0.7$)

$$(x - \mu_-)^2 = (0.7 - (-0.4667))^2 = (1.1667)^2 = 1.361$$

$$P(0.7 | -) = \frac{1}{\sqrt{2\pi(0.0422)}} \exp\left(-\frac{1.361}{2(0.0422)}\right)$$

Exponent term:

$$\frac{1.361}{0.0844} \approx 16.12$$
$$e^{-16.12} \approx 9.9 \times 10^{-8}$$

Denominator:

$$\sqrt{2\pi(0.0422)} \approx 0.515$$

So:

$$P(0.7 | -) = \frac{9.9 \times 10^{-8}}{0.515} \approx 1.9 \times 10^{-7}$$

STEP 5 — Compute Posterior Probabilities

We need:

$$P(\text{Positive} | x) \propto P(\text{Positive}) \cdot P(\text{No} | \text{Positive}) \cdot P(0.7 | \text{Positive})$$

$$P(\text{Negative} | x) \propto P(\text{Negative}) \cdot P(\text{No} | \text{Negative}) \cdot P(0.7 | \text{Negative})$$

Positive Posterior (unnormalized)

$$0.5 \cdot \frac{1}{3} \cdot 2.78$$
$$= 0.5 \cdot 0.333 \cdot 2.78 = 0.463$$

Negative Posterior (unnormalized)

$$0.5 \cdot \frac{2}{3} \cdot (1.9 \times 10^{-7})$$
$$= 0.5 \cdot 0.666 \cdot 1.9 \times 10^{-7} \approx 6.3 \times 10^{-8}$$

Hence +Ve

5. You are given the following small dataset of x and y: using a linear regression (a) least square error fit and (b) gradient descent 1 iteration with LR 0.05 to predicted value of y when x=6. **[10 +10 marks]**

x	y
1	2
2	5
3	7
4	10

Solution:

We use:

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

Step 1: Compute means

$$\bar{x} = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

$$\bar{y} = \frac{2 + 5 + 7 + 10}{4} = 6$$

x	y	$x - \bar{x}$	$y - \bar{y}$	Product
1	2	-1.5	-4	6
2	5	-0.5	-1	0.5
3	7	+0.5	+1	0.5
4	10	+1.5	+4	6

$$\sum (x - \bar{x})(y - \bar{y}) = 6 + 0.5 + 0.5 + 6 = 13$$

$$\sum (x - \bar{x})^2 = (-1.5)^2 + (-0.5)^2 + 0.5^2 + 1.5^2 = 2.25 + 0.25 + 0.25 + 2.25 = 5$$

$$b_1 = \frac{13}{5} = 2.6$$

$$b_0 = 6 - (2.6)(2.5) = 6 - 6.5 = -0.5$$

$$y = -0.5 + 2.6x$$



$$y = -0.5 + 2.6(6) = -0.5 + 15.6 = 15.1$$

(b)

We minimize:

$$J(b_0, b_1) = \frac{1}{2m} \sum_{i=1}^m (b_0 + b_1 x_i - y_i)^2$$

Initialize parameters

$$b_0 = 0, \quad b_1 = 0$$

Learning rate (α):

$$\alpha = 0.05$$

Number of iterations: 10 (just to illustrate)

Gradient formulas

$$b_0 := b_0 - \alpha \left(\frac{1}{m} \sum (b_0 + b_1 x_i - y_i) \right)$$

$$b_1 := b_1 - \alpha \left(\frac{1}{m} \sum (b_0 + b_1 x_i - y_i) x_i \right)$$

Iterative Computation (First Few Steps)

Iteration 1

Predictions (all zero initially):

Error terms = $[-2, -5, -7, -10]$

$$\frac{1}{4} \sum \text{errors} = -6$$

$$\frac{1}{4} \sum (\text{errors} \cdot x) = -18.25$$

Update:

$$b_0 = 0 - 0.05(-6) = 0.30$$

$$b_1 = 0 - 0.05(-18.25) = 0.925$$

Thus,

$$y = 0.30 + 0.925 \times 6 = 5.85$$

6. Given a binary classifier uses a logistic regression model:

[10 marks]

$$P(y = 1 | x) = \sigma(\beta_0 + \beta_1 x_1 + \beta_2 x_2)$$

You are given a small dataset of 4 observations:

Obs	x_1	x_2	y
1	2	1	1
2	1	3	0
3	3	2	1
4	2	4	0

perform one step of gradient ascent, using learning rate 0.1, starting from:

$$\beta_0 = 0, \quad \beta_1 = 0.5, \quad \beta_2 = -0.25$$

1. Compute predicted probabilities

Model:

$$p_i = \sigma(z_i), \quad z_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$$

Current parameters:

- $\beta_0 = 0$
- $\beta_1 = 0.5$
- $\beta_2 = -0.25$

Compute each z_i :

Obs 1:

$$z_1 = 0 + 0.5(2) - 0.25(1) = 1 - 0.25 = 0.75$$

$$p_1 = \sigma(0.75) = \frac{1}{1 + e^{-0.75}} = 0.679$$

Obs 2:

$$z_2 = 0 + 0.5(1) - 0.25(3) = 0.5 - 0.75 = -0.25$$

Obs 3:

$$z_3 = 0 + 0.5(3) - 0.25(2) = 1.5 - 0.5 = 1.0$$

$$p_3 = \sigma(1) = 0.731$$

Obs 4:

$$z_4 = 0 + 0.5(2) - 0.25(4) = 1 - 1 = 0$$

$$p_4 = \sigma(0) = 0.5$$

Predictions summary

Obs	p_i
1	0.679
2	0.438
3	0.731
4	0.500

2. Compute gradients

General gradient:

$$\frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^n (y_i - p_i) x_{ij}$$

Let residuals $r_i = y_i - p_i$:

- Obs 1: $1 - 0.679 = 0.321$
- Obs 2: $0 - 0.438 = -0.438$
- Obs 3: $1 - 0.731 = 0.269$
- Obs 4: $0 - 0.500 = -0.500$

Gradient for β_0 :

$$g_0 = \sum r_i = 0.321 - 0.438 + 0.269 - 0.500 = -0.348$$

Gradient for β_1 :

$$\begin{aligned} g_1 &= \sum r_i x_{1i} \\ &= 0.321(2) - 0.438(1) + 0.269(3) - 0.500(2) \\ &= 0.642 - 0.438 + 0.807 - 1.000 = 0.011 \end{aligned}$$

Gradient for β_2 :

$$\begin{aligned} g_2 &= \sum r_i x_{2i} \\ &= 0.321(1) - 0.438(3) + 0.269(2) - 0.500(4) \\ &= 0.321 - 1.314 + 0.538 - 2.000 = -2.455 \end{aligned}$$

Gradient summary

$$g_0 = -0.348, \quad g_1 = 0.011, \quad g_2 = -2.455$$

3. Gradient ascent update

$$\beta_j^{\text{new}} = \beta_j + \alpha g_j$$

Learning rate: $\alpha = 0.1$

Update each parameter:

β_0

$$\beta_0^{\text{new}} = 0 + 0.1(-0.348) = -0.0348$$

β_1

$$\beta_1^{\text{new}} = 0.5 + 0.1(0.011) = 0.5011$$

β_2

$$\beta_2^{\text{new}} = -0.25 + 0.1(-2.455) = -0.4955$$

4. Final Updated Parameters

$\beta_0 = -0.0348, \quad \beta_1 = 0.5011, \quad \beta_2 = -0.4955$
--

7. With least square error fit use the given data to predict the value for $x_1=1$, $x_2=2$. [10 marks]

Obs	x_1	x_2	y
1	2	1	4
2	3	4	10
3	5	2	12
4	7	5	20
5	9	7	25

We will compute:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Initial Matrices:

$$X = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 4 \\ 1 & 5 & 2 \\ 1 & 7 & 5 \\ 1 & 9 & 7 \end{bmatrix} \quad y = \begin{bmatrix} 4 \\ 10 \\ 12 \\ 20 \\ 25 \end{bmatrix}^T$$

1. Correct Matrices Setup:

$$X^T X = \begin{bmatrix} 5 & 26 & 19 \\ 26 & 168 & 122 \\ 19 & 122 & 95 \end{bmatrix} \quad X^T y = \begin{bmatrix} 71 \\ 463 \\ 343 \end{bmatrix}^T$$

$$\det(X^T X) = 1048$$

$$(X^T X)^{-1} = \frac{1}{1048} \begin{bmatrix} 1076 & -152 & -20 \\ -152 & 114 & -116 \\ -20 & -116 & 164 \end{bmatrix}$$

Multiply the matrices:

$$\hat{\beta} = \frac{1}{1048} \begin{bmatrix} 1076 & -152 & -20 \\ -152 & 114 & -116 \\ -20 & -116 & 164 \end{bmatrix} \begin{bmatrix} 71 \\ 463 \\ 343 \end{bmatrix}^T$$

Carrying out the multiplication:

- First element: $1076(71) - 152(463) - 20(343) = 76396 - 70376 - 6860 = -840$
- Second element: $-152(71) + 114(463) - 116(343) = -10792 + 52782 - 39788 = 2202$
- Third element: $-20(71) - 116(463) + 164(343) = -1420 - 53708 + 56252 = 1124$

Thus:

$$\hat{\beta} = \frac{1}{1048} \begin{bmatrix} -840 \\ 2202 \\ 1124 \end{bmatrix}^T$$

Divide individually:

$$\beta_0 = \frac{-840}{1048} \approx -0.8015$$

$$\beta_1 = \frac{2202}{1048} \approx 2.1011$$

$$\beta_2 = \frac{1124}{1048} \approx 1.0725$$

Final Regression Equation:

$$y = -0.8015 + 2.1011(x_1) + 1.0725(x_2)$$

Prediction for $x_1=1$, $x_2=2$:

$$y = -0.8015 + 2.1011(1) + 1.0725(2)$$

$$y = -0.8015 + 2.1011 + 2.1450$$

$$y \approx 3.44$$

8. Many patients visit clinics with symptoms like fever, cough, headache, and fatigue. Doctors often rely on experience to diagnose the disease, but inexperienced practitioners or situations with limited resources may lead to misdiagnosis. There is a need for an expert system that can assist

healthcare providers in diagnosing common diseases by analyzing patient symptoms and providing possible diagnoses along with recommendations for further tests or treatments. For the given set of data build an expert system rule base. [10 marks]

Rule	IF (Symptoms)	THEN (Disease)	Confidence Factor
1	Fever AND Headache AND Fatigue	Malaria	0.8
2	Fever AND Rash AND Joint Pain	Dengue	0.9
3	Fever AND Cough AND Sore Throat	Influenza	0.7
4	Fever AND Headache AND Nausea	Typhoid	0.85
5	Cough AND Sore Throat AND Mild Fever	Common Cold	0.6

```

if 'fever' in symptoms:
    if 'headache' in symptoms:
        if 'fatigue' in symptoms:
            disease = "Malaria"
            confidence = 0.8
            reason = "Patient has fever, headache, and fatigue."
        elif 'nausea' in symptoms:
            disease = "Typhoid"
            confidence = 0.85
            reason = "Patient has fever, headache, and nausea."
        else:
            disease = "Unknown"
            confidence = 0
            reason = "Symptoms partially match."
    elif 'rash' in symptoms and 'joint_pain' in symptoms:
        disease = "Dengue"
        confidence = 0.9
        reason = "Patient has fever, rash, and joint pain."
    elif 'cough' in symptoms and 'sore_throat' in symptoms:
        disease = "Influenza"
        confidence = 0.7
        reason = "Patient has fever, cough, and sore throat."
    else:
        disease = "Unknown"
        confidence = 0
        reason = "Fever present but other symptoms do not match."
    elif 'cough' in symptoms and 'sore_throat' in symptoms and 'mild_fever' in symptoms:
        disease = "Common Cold"
        confidence = 0.6
        reason = "Patient has cough, sore throat, and mild fever."
    else:
        disease = "Unknown"
        confidence = 0
        reason = "Symptoms do not match any known disease."

```

9. A small bookstore recorded weekly sales (in units) of a new book over 15 weeks: 20, 22, 25, 30, 2, 3, 5, 14, 15, 18, 6, 8, 9, 12 You are asked to: group the data into 4 unequal-width and overlapping bins and smooth by boundary and mean. [10 marks]

For binning there could be multiple solutions following is one of them. All valid solutions will, be considered

Sort the Data: 2, 3, 5, 6, 8, 9, 12, 14, 15, 18, 20, 22, 25, 30

We choose reasonable overlapping ranges:

Bin 1 (0–10):

2, 3, 5, 6, 8, 9

Bin 2 (5–15):

5, 6, 8, 9, 12, 14, 15, 18

Bin 3 (12–22):

12, 14, 15, 18, 20, 22

Bin 4 (20–35):

20, 22, 25, 30

Boundary Smoothed Data

2, 2, 2, 9, 9, 9

5, 5, 5, 18, 18, 18, 18, 18,

12, 22, 22, 22, 22, 22,

20, 20, 30, 30

Mean Smoothed Data

5.5, 5.5, 5.5, 5.5, 5.5, 5.5,

8.1875, 8.1875, 8.1875, 8.1875, 8.1875, 8.1875, 8.1875, 8.1875

13.85, 13.85, 13.85, 13.85, 13.85, 13.85

24.25, 24.25, 24.25, 24.25

10. Given the initial state of 4 queens, show the steps for the hill climbing algorithm until it terminates. Provide if multiple neighbours have the same cost, choose the neighbour where the pair of queens swapped has the smallest subscript/column number. For example, when we can swap either (x0,x4) or (x2,x3), we will swap (x0,x4). When we can swap either (x2,x3) or (x2,x4) we will swap (x2,x3). [15 marks]

			Q
	Q		
		Q	
Q			

Both x0-3/ x1-4 will consider as notation for answer provided steps are correct

Solution: Neighbours of $x_0 = 3, x_1 = 1, x_2 = 2, x_3 = 0$ (cost is 2.)

- $x_0 = 1, x_1 = 3, x_2 = 2, x_3 = 0$ cost is 1. Choose this neighbour.
- $x_0 = 2, x_1 = 1, x_2 = 3, x_3 = 0$ cost is 1.
- $x_0 = 0, x_1 = 1, x_2 = 2, x_3 = 3$ cost is 6.
- $x_0 = 3, x_1 = 2, x_2 = 1, x_3 = 0$ cost is 6.
- $x_0 = 3, x_1 = 0, x_2 = 2, x_3 = 1$ cost is 1.
- $x_0 = 3, x_1 = 1, x_2 = 0, x_3 = 2$ cost is 1.

Neighbours of $x_0 = 1, x_1 = 3, x_2 = 2, x_3 = 0$ (cost is 1.)

- $x_0 = 3, x_1 = 1, x_2 = 2, x_3 = 0$ cost is 1.
- $x_0 = 2, x_1 = 3, x_2 = 1, x_3 = 0$ cost is 2.
- $x_0 = 0, x_1 = 3, x_2 = 2, x_3 = 1$ cost is 4.
- $x_0 = 1, x_1 = 2, x_2 = 3, x_3 = 0$ cost is 4.
- $x_0 = 1, x_1 = 0, x_2 = 2, x_3 = 3$ cost is 2.
- $x_0 = 1, x_1 = 3, x_2 = 0, x_3 = 2$ cost is 0. Choose this neighbour.